Camera QC Report

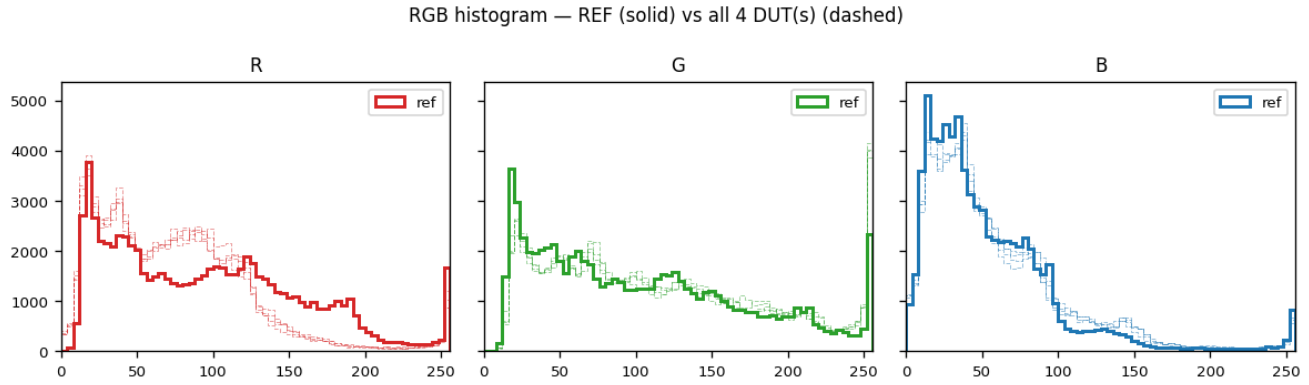
FAIL — DUT is worse than reference on multiple metrics.

- **Generated:** 2026-06-22 14:38 UTC
- **Scene preset:** `chessboard`
- **Reference:** `reference.jpg`
- **DUT images:** 4

Disclaimer — preliminary thresholds. The pass/marginal/fail thresholds used in this report are preliminary and should be tuned against a batch of known-good DUTs before this report is used as a standalone acceptance decision.

Summary (per metric, across all DUT images)

Metric	Direction	Tol %	Passed
L	closer to ref	10.0	4 / 4
a*	closer to ref	7.0	0 / 4
b*	closer to ref	12.0	4 / 4
R	closer to ref	10.0	0 / 4
G	closer to ref	10.0	2 / 4
B	closer to ref	7.0	0 / 4
sharpness	higher better	10.0	0 / 4
noise	lower better	0.0	4 / 4
sharpness_fft	higher better	10.0	3 / 4



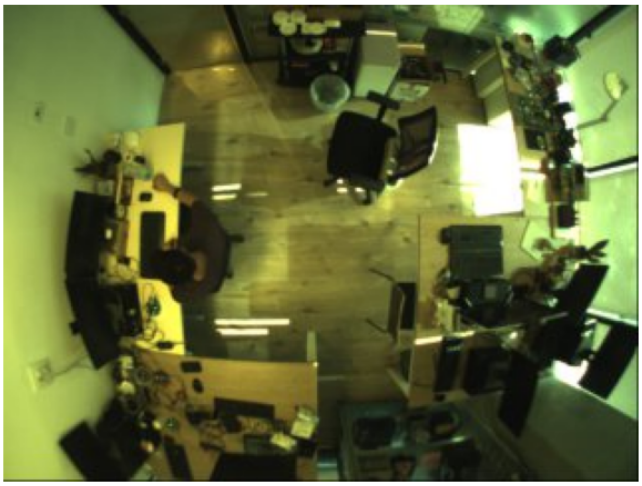
Direction key: *higher better* — DUT failing means it dropped >tol% below ref. *lower better* — DUT failing means it rose >tol% above ref. *closer to ref* — DUT failing means |Δ| exceeds tol% of ref.

Per-DUT-image comparisons

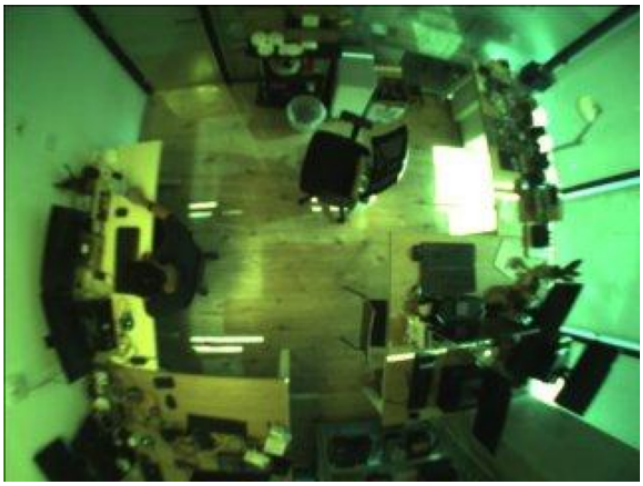
DUT 1: `0.jpg`

FAIL — two or more metrics outside tolerance. (4/9 metrics outside tolerance)

REF: reference.jpg



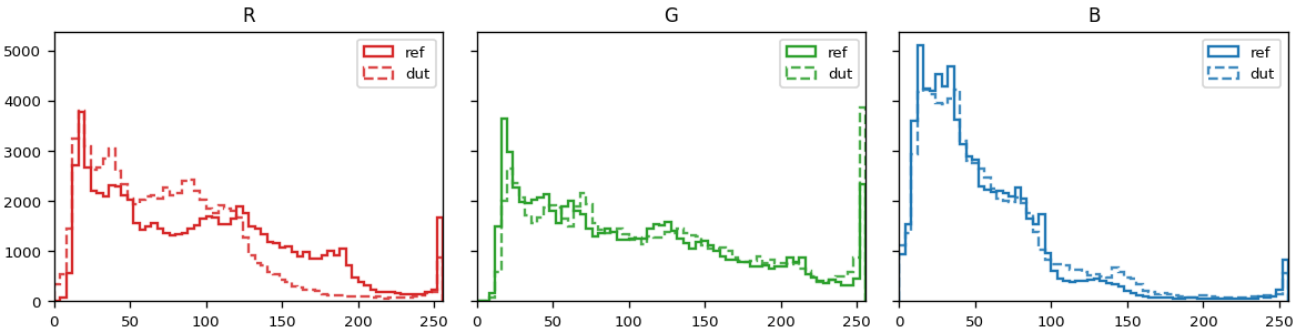
DUT: 0.jpg



Conclusion: sharpness 85.7 / 90.2% (avg ~88%); noise 91% of ref (cleaner); channels mixed (-23.1 to +9.2%); tint shift a*=-12.79, b*=-0.49.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	95.44	94.97	-0.47	99.5	10.0	OK
a*	-10.68	-23.47	-12.79	219.7	7.0	OUT
b*	25.44	24.95	-0.49	98.1	12.0	OK
R	95.31	73.27	-22.04	76.9	10.0	OUT
G	103.52	113.08	9.56	109.2	10.0	OK
B	54.11	58.58	4.47	108.3	7.0	OUT
sharpness	794.45	680.54	-113.91	85.7	10.0	OUT
noise	9.31	8.51	-0.80	91.4	0.0	OK
sharpness_fft	2617.84	2361.82	-256.03	90.2	10.0	OK

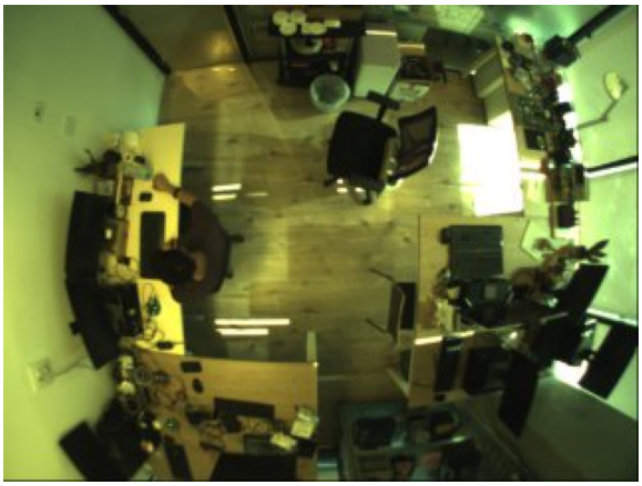
RGB histogram — REF (solid) vs DUT (dashed)



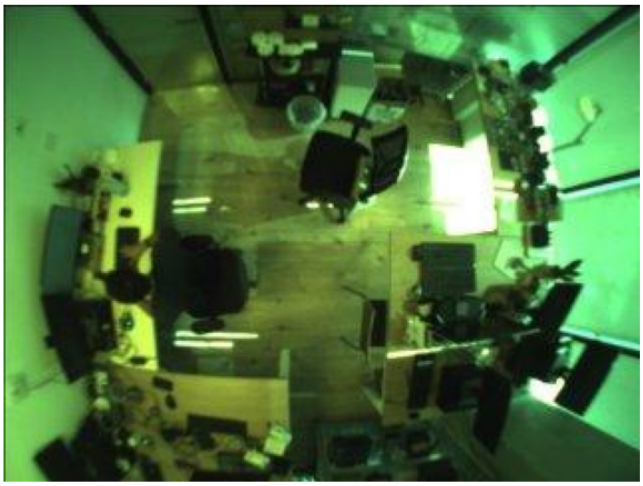
DUT 2: `1.jpg`

FAIL — two or more metrics outside tolerance. (4/9 metrics outside tolerance)

REF: reference.jpg



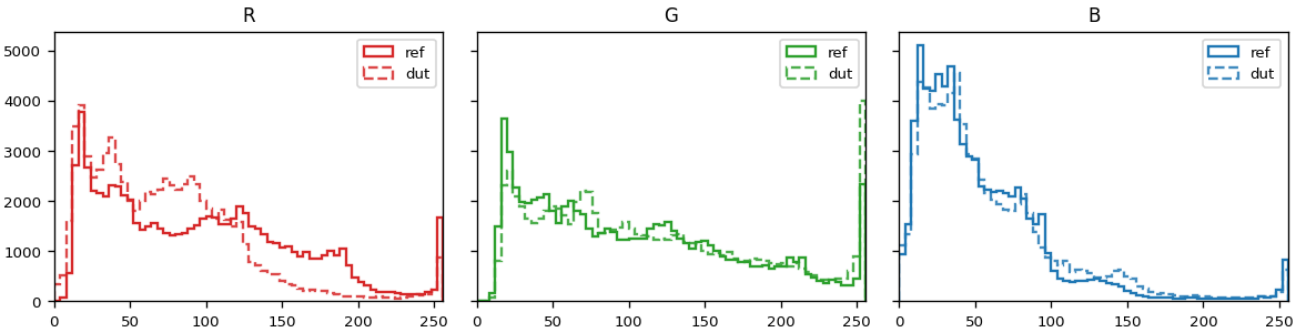
DUT: 1.jpg



Conclusion: sharpness 87.1 / 90.9% (avg ~89%); noise 92% of ref (cleaner); channels mixed (-24.1 to +8.5%); tint shift a*=-12.77, b*=-0.90.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	95.44	94.25	-1.19	98.8	10.0	OK
a*	-10.68	-23.45	-12.77	219.6	7.0	OUT
b*	25.44	24.54	-0.90	96.5	12.0	OK
R	95.31	72.30	-23.01	75.9	10.0	OUT
G	103.52	112.34	8.82	108.5	10.0	OK
B	54.11	58.64	4.53	108.4	7.0	OUT
sharpness	794.45	692.10	-102.35	87.1	10.0	OUT
noise	9.31	8.60	-0.71	92.4	0.0	OK
sharpness_fft	2617.84	2378.90	-238.94	90.9	10.0	OK

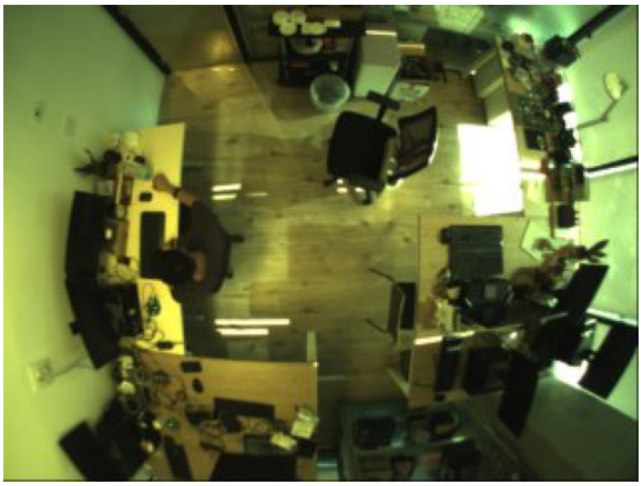
RGB histogram — REF (solid) vs DUT (dashed)



DUT 3: `2.jpg`

FAIL — two or more metrics outside tolerance. (5/9 metrics outside tolerance)

REF: reference.jpg



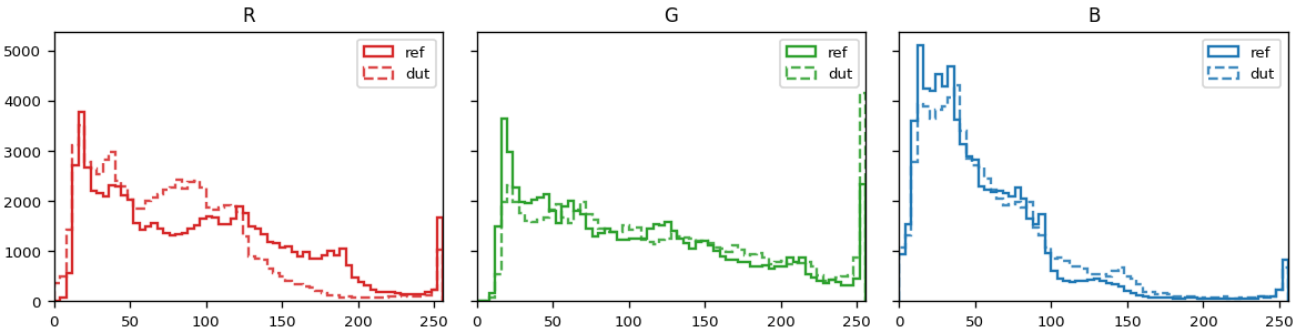
DUT: 2.jpg



Conclusion: sharpness 89.5 / 92.1% (avg ~91%); noise 94% of ref (cleaner); channels mixed (-21.0 to +12.6%); tint shift a*=-13.55, b*=+0.24.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	95.44	97.80	2.36	102.5	10.0	OK
a*	-10.68	-24.23	-13.55	226.8	7.0	OUT
b*	25.44	25.68	0.24	101.0	12.0	OK
R	95.31	75.29	-20.03	79.0	10.0	OUT
G	103.52	116.56	13.03	112.6	10.0	OUT
B	54.11	60.26	6.15	111.4	7.0	OUT
sharpness	794.45	710.96	-83.49	89.5	10.0	OUT
noise	9.31	8.74	-0.57	93.9	0.0	OK
sharpness_fft	2617.84	2412.19	-205.65	92.1	10.0	OK

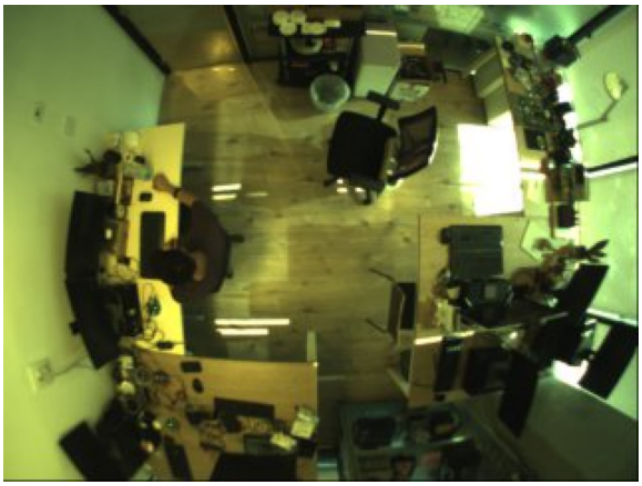
RGB histogram — REF (solid) vs DUT (dashed)



DUT 4: `3.jpg`

FAIL — two or more metrics outside tolerance. (6/9 metrics outside tolerance)

REF: reference.jpg



DUT: 3.jpg



Conclusion: sharpness 83.4 / 88.1% (avg ~86%); noise 90% of ref (cleaner); channels mixed (-22.2 to +10.9%); tint shift a*=-13.18, b*=-0.19.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	95.44	96.32	0.88	100.9	10.0	OK
a*	-10.68	-23.86	-13.18	223.4	7.0	OUT
b*	25.44	25.25	-0.19	99.2	12.0	OK
R	95.31	74.13	-21.18	77.8	10.0	OUT
G	103.52	114.77	11.24	110.9	10.0	OUT
B	54.11	59.51	5.40	110.0	7.0	OUT
sharpness	794.45	662.43	-132.02	83.4	10.0	OUT
noise	9.31	8.34	-0.97	89.6	0.0	OK
sharpness_fft	2617.84	2307.34	-310.51	88.1	10.0	OUT

RGB histogram — REF (solid) vs DUT (dashed)

